



7686 - Brown Dwarf Broiler: Probing Chemical Quenching and Heat Redistribution in a Highly-Eccentric Brown Dwarf

Cycle: 4, Proposal Category: GO

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Elijah Mullens (PI)	Cornell University
Dr. Ryan Challener (CoI) (CoPI) (Contact)	Cornell University
Dr. Nikole Lewis (CoI) (US Admin CoI) (Contact)	Cornell University
Maura Lally (CoI)	Cornell University
Victoria Abigail Boehm (CoI)	Cornell University
Isabela Grace Huckabee (CoI)	Cornell University
Dr. Channon Visscher (CoI)	Space Science Institute
Dr. Ian Crossfield (CoI)	University of Kansas Center for Research, Inc.
Dr. Tiffany Kataria (CoI)	Jet Propulsion Laboratory
Dr. Genaro Suarez (CoI)	American Museum of Natural History
Dr. Johanna Vos (CoI) (ESA Member)	Trinity College Dublin
Dr. Eileen Gonzales (CoI)	San Francisco State University
Dr. Jacqueline Kelly Faherty (CoI)	American Museum of Natural History
Dr. Brianna Irene Lacy (CoI)	University of California - Santa Cruz

OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Partial Phase Curve				
	1		NIRSpec Bright Object Time Series	(1) TYC-7052-1753-1

ABSTRACT

JWST observations have begun to shed light on insufficiencies of the chemical networks and dynamical models we currently use to understand highly irradiated exoplanets and isolated brown dwarfs. Eccentric, high mass, transiting targets allow for a nature-given laboratory to study how interconversion timescales vary with changing levels of incident stellar flux, and therefore allow us to disentangle which processes dominate in high-mass substellar object atmospheres. TOI 2490 b is the most eccentric transiting brown dwarf in the brown dwarf desert and provides a high-mass comparison object to the highly eccentric hot Jupiter HD 89060 b, allowing us to bridge the exoplanet and brown dwarf population.

Our program will observe a partial phase-curve (26.385 hours total) using the NIRSpec G395H grism to obtain spectra from 2.87-5.27 microns, including both periastron passage and secondary eclipse. This wavelength range encompasses strong spectral features of the key carbon-bearing species (CH_4 , CO_2 , and CO) that are tracers of the physical and chemical processes, such as chemical quenching, we aim to track. The precision of G395H allows us to track their abundances over our time-series observation and test the favored reaction pathways for CH_4 - CO interconversion, and how much chemical timescales lag behind heating timescales. By observing secondary-eclipse, we will probe dynamical mixing and heat redistribution through eclipse mapping.

OBSERVING DESCRIPTION

We will observe a partial phase-curve (26.385 hours = 1 hour scheduling window + 2 hours before periastron passage + 18.355 hours from periastron passage to eclipse + 3.03 hour eclipse + 2 hours after eclipse) using the NIRSpec G395H grism in BOTS mode to obtain spectra from 2.87-5.27 microns, including both periastron passage and secondary eclipse. We will collect 70 groups per integration with 1483 integrations covering the entire duration. When including overheads, this results in a total charge time of 31.7 hours.

Phases were computed with ExoCTK with $P = 60.3326$ days, $e = 0.779$, $w = 214.48$ degrees, inclination = 89.2, target observation = secondary eclipse. Time spent prior to mid eclipse was set to 21.87 hours (from mid eclipse to beginning of periastron passage + two hours before periastron passage). We have a one-hour scheduling window.

For NIRSpec WATA target acquisition (TA) using the F110W filter, TOI~2490 will saturate the central pixel. Previous studies of the comparably-bright $J \sim 10.663$ star WASP-39 encountered similar saturation using the same TA configuration, but regardless succeeded in acquiring directly on WASP-39 (PID 1366). We therefore will acquire directly on TOI 2490 through the F110W filter.

Proposal 7686 - Targets - Brown Dwarf Broiler: Probing Chemical Quenching and Heat Redistribution in a Highly-Eccentric Brown Dwarf

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	TYC-7052-1753-1	RA: 04 52 29.8573 (73.1244054d)	Proper Motion RA: 19.45 mas/yr	
			Dec: -36 15 25.75 (-36.25715d)	Proper Motion Dec: 12.607 mas/yr	
			Equinox: J2000	Parallax: 0.0037362000000000003"	
				Epoch of Position: 2000	
	<i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.</i> <i>Category=Star</i> <i>Description=[Brown dwarfs, Exoplanet Systems, Exoplanets, G dwarfs]</i> <i>Extended=NO</i>				

Proposal 7686 - Observation 1 - Brown Dwarf Broiler: Probing Chemical Quenching and Heat Redistribution in a Highly-Eccentric Bro...

Proposal 7686, Observation 1										Tue Apr 01 23:01:10 GMT 2025		
Observation	Diagnostic Status: Warning											
	Observing Template: NIRSpec Bright Object Time Series											
	Comments: Phases were computed with ExoCTK with P = 60.3326 days, e = 0.779, w = 214.48 degrees, inclination = 89.2, target observation = secondary eclipse, BJD of transit midpoint =2459180.6943 . Time spent prior to mid eclipse was set to 21.87 hours (from mid eclipse to beggining of periapse passage + two hours before periapse passage). We have a one hour scheduling window.											
	Full science observation is: 26.385 hours = 1 hour scheduling window + 2 hours before periastron passage + 18.355 hours from periastron passage to eclipse + 3.03 hour eclipse + 2 hours after eclipse											
For NIRSpec WATA target acquisition (TA) using the F110W filter, TOI~2490 will saturate the central pixel. Previous studies of the comparably-bright J~10.663 star WASP-39 encountered similar saturation using the same TA configuration, but regardless succeeded in acquiring directly on WASP-39 (PID 1366). We therefore will acquire directly on TOI 2490 through the F110W filter.												
Diagnostics	(Observation 1) Warning (Form): Exposure Duration exceeds the limit of 10000.0 seconds. Above this limit it is possible that a High Gain Antenna move may occur during the exposure.											
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	TYC-7052-1753-1	RA: 04 52 29.8573 (73.1244054d) Dec: -36 15 25.75 (-36.25715d) Equinox: J2000				Proper Motion RA: 19.45 mas/yr Proper Motion Dec: 12.607 mas/yr Parallax: 0.0037362000000000003" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Brown dwarfs, Exoplanet Systems, Exoplanets, G dwarfs] Extended=NO											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	SAME	WATA	SUB32	F110W	NRSRAPID	3	1	1	0.08	221788	
Template	Subarray											
	SUB2048											
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID		
	1	G395H/F290LP	NRSRAPID	70	1483	1	1	1483	95004.658	221788		

Proposal 7686 - Observation 1 - Brown Dwarf Broiler: Probing Chemical Quenching and Heat Redistribution in a Highly-Eccentric Bro...

Special Requirements	Phase 0.07127307230699446 to 0.0719636884360303 with period 60.3326 Days and zero-phase 2459180.6943 HJD Time Series Observation No Parallel Attachments
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